INCH-POUND
MIL-DTL-3720F
27 November 2024
SUPERSEDING
MIL-S-3720E
2 December 1986

DETAIL SPECIFICATION

SKETCHING BOARD, WITH WATERPROOF COVER AND TRIPOD MOUNTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a wooden sketching board with built-in compass, a waterproof cover and a folding extension tripod.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MILITARY

MIL-T-704 - Treatment and Painting of' Materiel.

DEPARTMENT OF DEFENSE STANDARDS

MILITARY

MIL-STD-889 - Dissimilar Metals.

(Copies of these documents are available online at https://assist.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue Philadelphia, PA 19111-5096 or e-mail to CE Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of this documents are available online at https://www.asq.org or the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-B36/B36M	- Plate, Brass, Sheet, Strip, and Rolled Bar
ASTM-B121/B121M	- Plate, Leaded Brass, Sheet, Strip, and Rolled Bar
ASTM-B139/B139M	- Rod, Phosphor Bronze, Bar, and Shapes
ASTM-B209/B209M	- Aluminum and Aluminum-Alloy Sheet and Plate.
ASTM-B221	- Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM-B271/B271M	- Copper-Base Alloy Centrifugal Castings
ASTM-B316/B316M	- Aluminum and Aluminum-Alloy Rivet and Cold-Heading Wire and Rods.
ASTM-B584	- Copper Alloy Sand Castings for General Applications
ASTM-D1784	- Compounds and Chlorinated Poly (Vinyl Chloride) (CPVC) Compounds, Rigid Poly (Vinyl Chloride) (PVC)
ASTM-D4317	- Adhesives, Emulsion, Polyvinyl Acetate-Based
ASTM-D4444	- Meters, Hand-Held Moisture, Use and Calibration of

(Copies of these documents are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The sketching board, with waterproof cover and tripod (hereinafter specified as sketching board), shall be as shown on figure 1 through 4 and as specified herein.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2 and 6.3).

3.3 Material. Material shall be as specified herein and on the applicable figures. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The sketching board shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the sketching boards may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific

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material, material finish or treatment for use with component and subcomponent and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. Unless otherwise specified (see 3.3.11), the components pieces and parts incorporated in the sketching board may be newly fabricated from recovered materials to the maximum extent practicable, provided the sketching board produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the sketching board.

3.3.3 Cotton duck. Cotton duck shall be 12-ounce, hard textured, cotton cloth, treated for mildew resistance with copper 8-quinolinolate, and dyed green 383 in accordance with MIL-T-704. The treatment and dying processes shall not· leave material on the duck surface that will stain or adversely affect drawing paper carried in the cover fabricated from the duck.

3.3.4 Thread. Thread shall be three-cord (stitching) type made from long-line linen, treated for mildew resistance with copper 8-quinolinolate, and dyed green 383 in accordance with MIL-T-704.

3.3.5 Binding tape. Tape shall be plain weave general purpose cotton textile dyed and treated the same as cotton duck.

3.3.6 Closure tapes. The pile tape shall be nylon, 1 inch wide, 8.0 mil hook (330 denier), color black, and the bonded hook tape shall be nylon hook, nylon ground, 8.0 mil hook (330 denier), color black, Velcro brand or equal.

3.3.7 Phosphor bronze. Trie phosphor bronze used in the fabrication of the sketching board shall conform to ASTM-B139/B139M and, unless otherwise specified (see 6.2), composition and temper are optional.

3.3.8 Brass. The brass shall be leaded and nonleaded yellow brass, and shall conform to ASTM-B36/B36M or ASTM-B121/B121M, composition and temper optional, or ASTM-B271/B271M or ASTM-B584, Alloy No. c85400.

3.3.9 Aluminum. When specified (see 6.2), aluminum alloy 2024, temper T4 conforming to ASTM-B209/B209M, ASTM-B221 or ASTM-B316/B316M, may be substituted for phosphor bronze or brass materials.

3.3.10 Adhesive. The adhesive shall conform to ASTM-D4317.

3.3.11 Plastic molding material. The plastic molding material shall conform to ASTM-D1784, designation 14334, color natural.

3.3.12 Wood.

3.3.12.1 Softwood. Softwood shall be basswood, sugar pine, white pine, or ponderosa pine. The boards shall be free from defects, except that the bottom surface may contain mineral streaks. Sound knots of not more than 1/4-inch (6.4 millimeters (mm)) diameter and barely perceptible checks not more than 2 inches (50.8 mm) in length are acceptable. Light stain and mineral streaks occupying not more than 10 percent of the surface on which they appear shall not be considered defects. Moisture content of wood shall be not less than 6 percent nor more than 8 percent.

3.3.12.2 Hardwood. Hardwood shall be beech; birch, hickory, ash or pecan, and the slope of grain shall be not more than 1 in 15. The wood of the species used shall weigh not less than 40 pounds per cubic foot and shall be free from defects except for barely perceptible seasoning checks not more than 1 inch long and light stain and mineral streaks occupying not more than 10 percent of the surface on which they appear. Moisture content of wood shall be not less than 6 percent nor more than 8 percent.

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3.4 Detailed requirements.

3.4.1 Graduations, numbers, and markings. Graduations, numbers, and marking shall be as shown on figure 2 and as specified herein. Graduations, numbers, and marking shall be filled with a permanent black filler or made by screening or photoengraving, shall be uniform and distinct, and shall cause no bulging or chipping of the remaining surface.

3.4.2 Compass. The compass shall be induction damped. ·The north-seeking half of the compass trough shall contain a clear plastic for viewing the north- seeking end of the needle. The needle shall contain a movable balance weight. The top of the compass shall be flush with the top•·of the sketching board.

3.4.3 Tripod. The upper struts, when detached from the tripod head, shall fold so as to lay along the middle struts. The middle struts shall be rigid and shall contain a clamp-crew device for locking the lower struts in place when extended or allowing the lower struts to slide inside the middle struts.

3.4.4 Waterproof cover (stitching). The edge of the sketching board cover shall be sewed with the three-cord (stitching) type of long-line linen thread to provide a bound seam edge.

3.5 Dimensions. Dimensions of the sketching board and its components shall be as shown on figures 1 through 4.

3.6 Color and finish.

3.6.1 Sketching board. The board shall be sanded and given two coats of spar varnish. The brass fittings shall be provided a black chemical finish produced by chemical or electrochemical treatment of the copper-alloy surfaces.

3.6.2 Tripod. The tripod legs shall be sanded, treated, and painted in accordance with MIL-T-704, type A, to provide protection against rust, corrosion, and deterioration. Only one coat of primer shall be applied. Metal fittings, except for threaded parts and bearing surfaces shall be given a durable, lusterless finish, color approximating green 383 of MIL-T-704, type F or G, as applicable.

3.7 Marking.

3.7.1 Identification marking. The sketching board identification shall include the following: Item name, stock number, procurement instrument identification number, and the symbol "US". The marking shall be located on the back of the sketching board to the right of the instruction marking specified in 3.7.2.

3.7.2 Instruction marking. The following marking shall appear on the back of the sketching board in the center between the trough and the screw plate:

INSTRUCTIONS (1/4 in. high letters)
(0.2-inch-high letters for 1 through 7)

1. STORE BOARD FLAT WITH THIS SIDE UP.
2. USE ONLY CLAMPS TO ATTACH PAPER.
3. SCREW BOARD LIGHTLY TO TRIPOD.
4. SET UP TRIPOD TO GIVE APPROXIMATE ORIENTATION.
5. MAKE FINAL ORIENTATION BY TURNING BOARD ON TRIPOD.
6. AFTER FINAL ORIENTATION. SCREW BOARD FIRMLY ON TRIPOD.
7. NEVER USE FORCE TO MAKE ANY ADJUSTMENT.

3.8 Workmanship.

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3.8.1 Sketching board and tripod. Sketching boards and tripods shall be clean and free from warpage, chipping, seaming, or splintering. All screws shall be fully turned down. All metal parts shall be clean and free from burrs and harmful extraneous material. All edges shall be rounded or beveled. The struts shall fold and slide without twisting, binding, or bending.

3.8.2 Waterproof cover. Finished covers shal1 be clean and free from abrasion marks permanently embedded creases, loopy or loose edges, clearly visible dye stains or streaks, and oil spots created by treating which would affect appearance.

4. VERIFICATION

4.1 Classification of inspections. Inspection requirements specified herein are classified As follows:

a. First article inspection (see 4.2).
b. Quality conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 Examination. The first article sketching board, cover, and tripod shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.2.2 Tests The first article sketching board, cover, and tripod shall be tested as specified in 4.4.2. Failure of the test shall be cause for rejection.

4.3 Conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with ASQ Z1.4.

4.3.2 Examination. Samples s elected in accordance with 4.3.1 shall be examined as specified in 4.4.1.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.

4.4.1 Examination. The sketching board, cover, and tripod shall be examined as specified herein for the defects listed in table I.

4.4.2 Tests. The moisture content of the wood in the sketching board; tripod, and tripod head shall be tested in accordance with ASTM-D4444. A moisture content less than 6 percent or more than 8 percent shall constitute failure of this test.

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TABLE I. Examination Schedule

Number	Defects	Requirement Paragraph
Major		
101.	Material not as specified. Each incorrect material shall constitute one defect.	3.3
102.	Materials are not resistant to corrosion and deterioration or treated to be-resistant to corrosion and deterioration for the applicable and operating environments.	3.3.1
103.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.1.1
104.	Contractor does not have documentation available for identification of material, material finishes or treatment.	3.3.2
105.	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the sketching board.	3.3.2
106.	Graduations and numbering not as Specified.	3.4.1
107	Compass not as specified	3.4.2
108.	Tripod not as specified	3.4.3
109.	Seam type not as Specified	3.4.4
110.	Stitch type not as required	3.4.4
111.	Dimensions not as Specified	3.5
112.	Workmanship not as Specified	3.8
Minor		
201.	Color and finish not as specified	3.6
202.	Marking not as specified	3.7

5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirement shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsibility packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control point's packaging activity within the Military department of defense Agency or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

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(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The case is intended for carrying engineer battalion drafting equipment used in field reconnaissance.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required (see 3.1, 4.2 and 6.3)
c. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the items should be an initial production. The first article should consist of one or more units. The contracting Officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first articles

6.4 Subject term key words listing.

Aluminum alloy 2024
Basswood
Beech
Brass
Built-in-compass.
Hickory
Pecan
Phosphor bronze
Ponderosa pine
Sugar pine
Tripod
White pine
Wooden sketching board

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - CE
Air Force - 71
DLA - CQ

Reviewer:
Navy - MC

Preparing Activity:
DLA – CQ

(Project 6675-2024-002)

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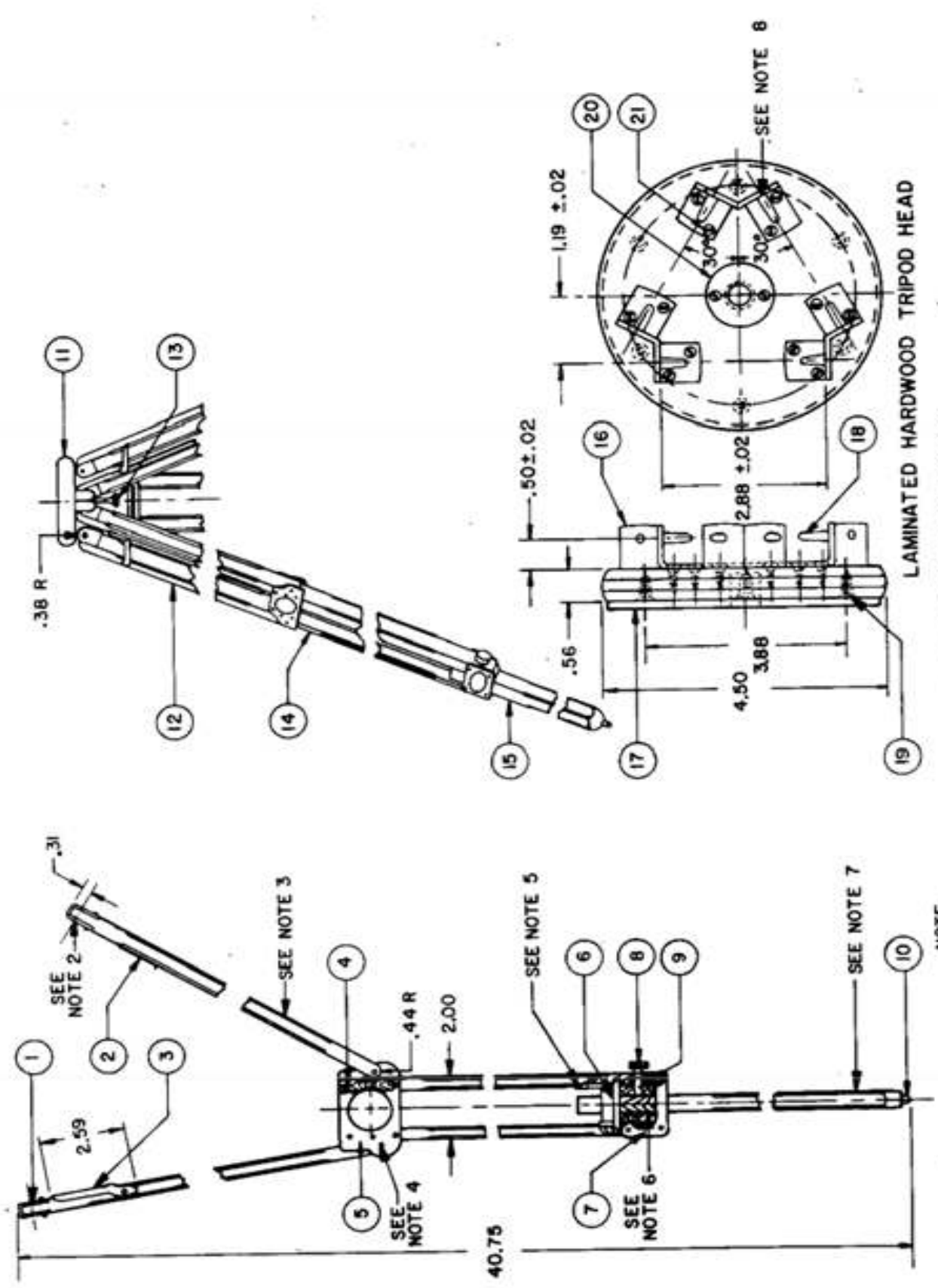


FIGURE 1A. Tripod and tripod head details.

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FIND NO.	QTY.	DESCRIPTION AND SPECIFICATION
1	6	BRASS END CLIP, NO. 20 GAGE (B&S)
2	3	BRASS SPREADER STOP, NO. 20 GAGE (B&S)
3	3	BRASS SPREADER, NO. 16 GAGE (B&S)
4	36	BRASS RIVET, .06±.02DIA ROUND HEAD
5	6	BRASS UPPER TIE PLATE, NO. 20 GAGE (B&S)
6	6	PHOSPHOR BRONZE SLIDING PIN
7	6	BRASS LOWER TIE PLATE, NO. 20 GAGE (B&S)
8	3	BRASS SLEEVE NUT (KNURL)
9	3	BRASS CLAMP SCREW, 10-32 UNF-1A
10	3	STEEL PIN
11	1	LAMINATED HARDWOOD TRIPOD HEAD
12	6	UPPER STRUT, WOOD, .44 BY .81 BY 14.00
13	1	BRASS TRIPOD SCREW .375-16 UNC-1A (CAPTIVE)
14	6	MIDDLE STRUT, WOOD, .44 BY .81 BY 15.00
15	3	LOWER STRUT, WOOD .56 BY .81 BY 14.31
16	3	BRASS LEG CLIP NO. 14 GAGE (B&S)
17	1	FELT CUSHION .06±.02 TH BY 4.31±.03 DIA
18	6	BRASS PIN .56±.02 LONG BY .19±.02 TAPER TO .12±.02
19	6	BRASS SCREW, FLAT HEAD, NO. 7 BY .75 (NIP OFF POINTS
		AFTER SCREWS ARE DRAWN UP TIGHTLY).
20	1	BRASS PLATE 1.12±.02 DIA BY .09 ± .02 TH CENTER TAPPED
		.375-16 UNC-1B
21	12	BRASS SCREW, FLAT HEAD, NO. 5 BY .50

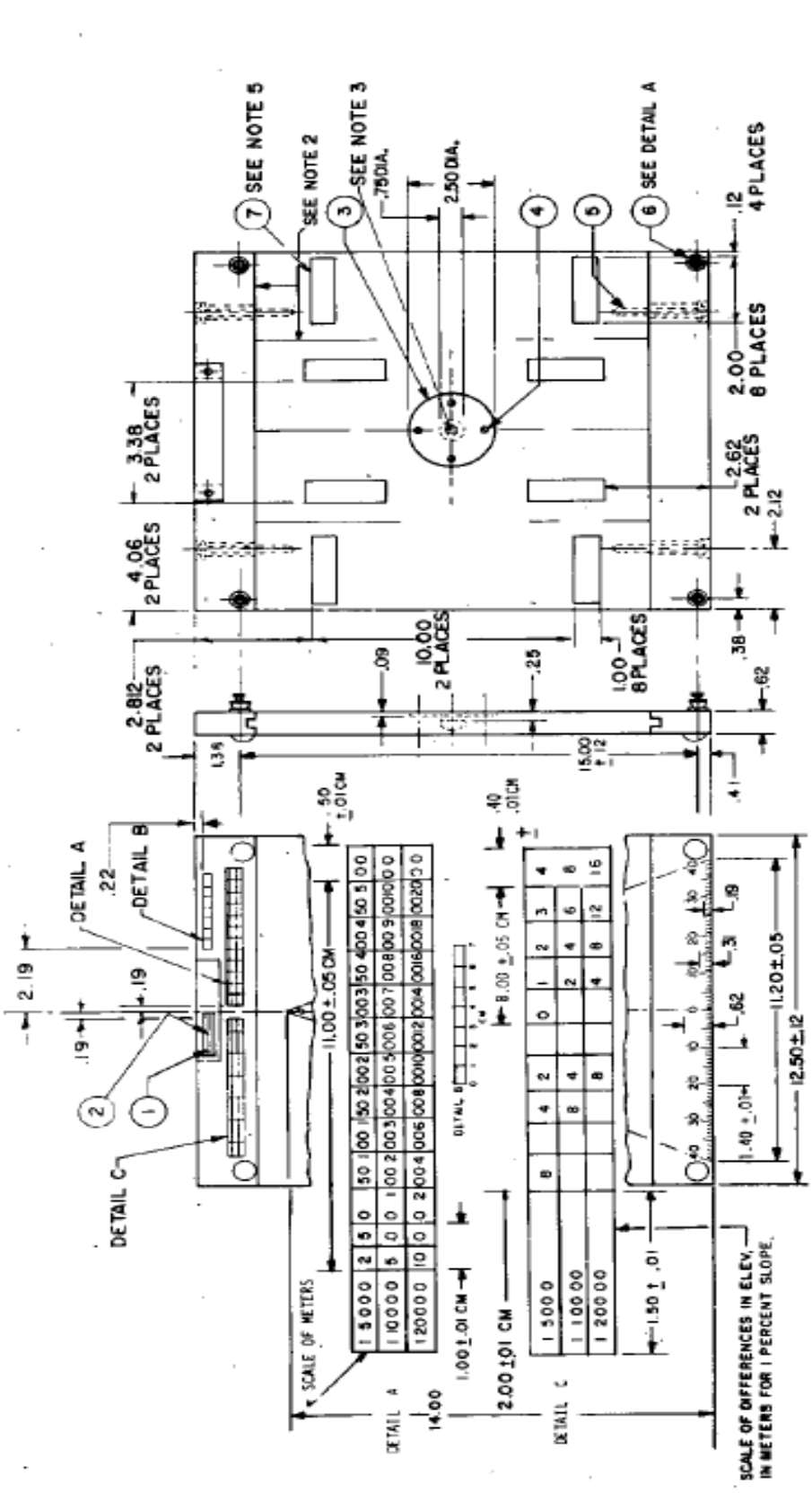
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DRILL .16 ± .02 DIA AT ASSEMBLY.
3. CHAMFER .06 BY 45°.
4. PIVOT POINT FOR FOLDING UPPER STRUT ONTO MIDDLE STRUT.
5. CHANNEL FOR SLIDEPIN.
6. GLUE HARDWOOD SPACERS TO MIDDLE STRUT
7. LOWER STRUT SLIDES INTO MIDDLE STRUT.
8. BRASS PIN TO BE RIVETED IN PLACE.
9. UNLESS OTHERWISE SPECIFIED TOLERANCE SHALL BE ± .03.

FIGURE 1B. Tripod and tripod head details.

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FIND NO	QTY.	DESCRIPTION AND SPECIFICATION
1	1	PLASTIC WINDOW, .06 THK.
2	1	COMPASS
3	1	BRASS SCREW PLATE, CENTER TAP, .375-16 UNC-1B.
4	4	BRASS SCREW, WOOD FL HD, NO.6 BY .50
5	4	BRASS SCREW, WOOD RD HD, NO.10 BY 2.50.
6	4	CLAMP PAPER (SEE FIG 3 FOR DETAILS).
7	8	BONDED HOOK TAPE

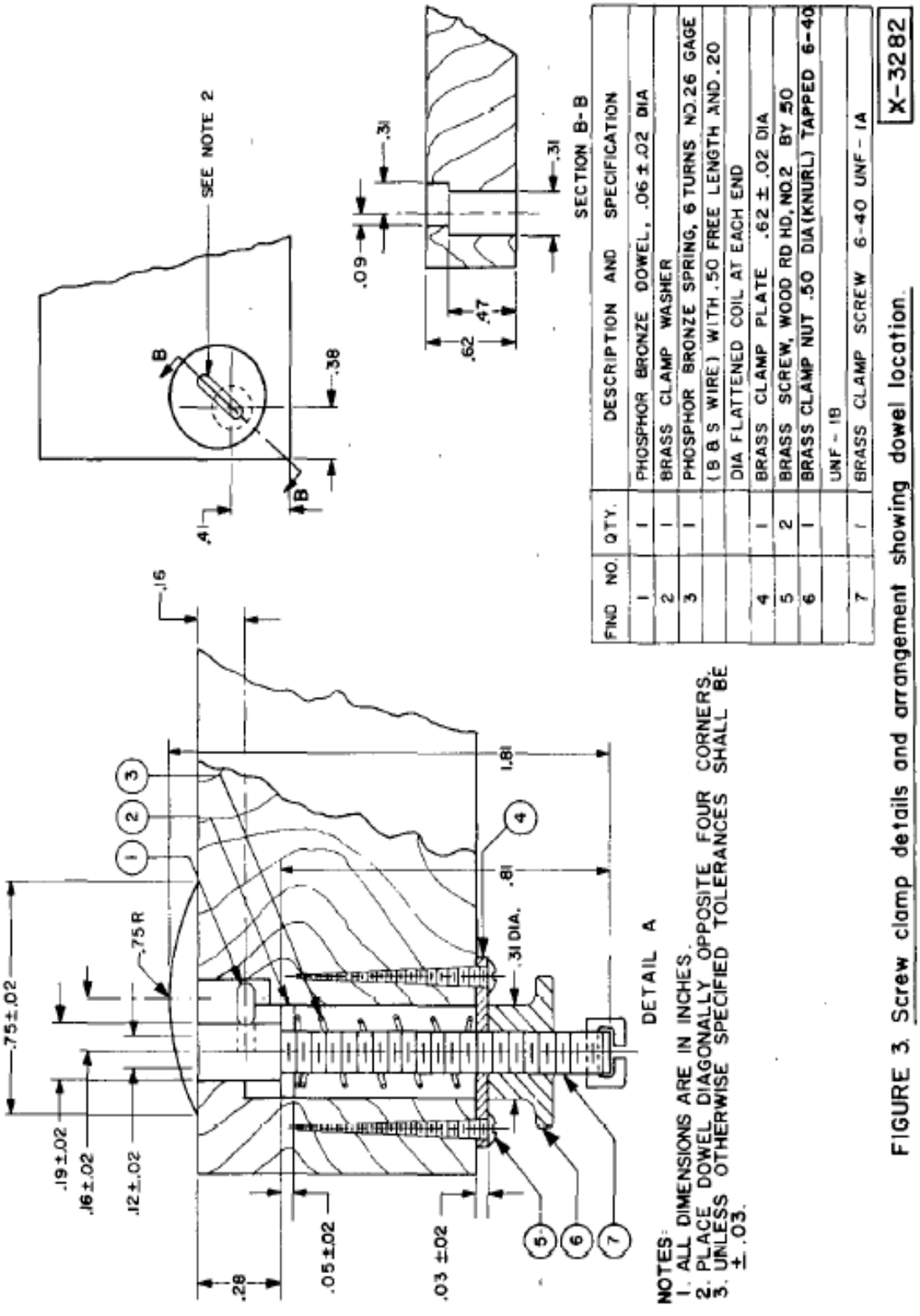
NOTES:

1. UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES.
2. GLUED TONGUE AND GROOVE JOINTS.
3. CENTER ON BOARD WITHIN .12.
4. UNLESS OTHERWISE SPECIFIED TOLERANCES SHALL BE ± .03.
5. BOND HOOK TAPE, FIND NO. 7, TO BACK OF SKETCH BOARD.

FIGURE 2. Sketching board details.

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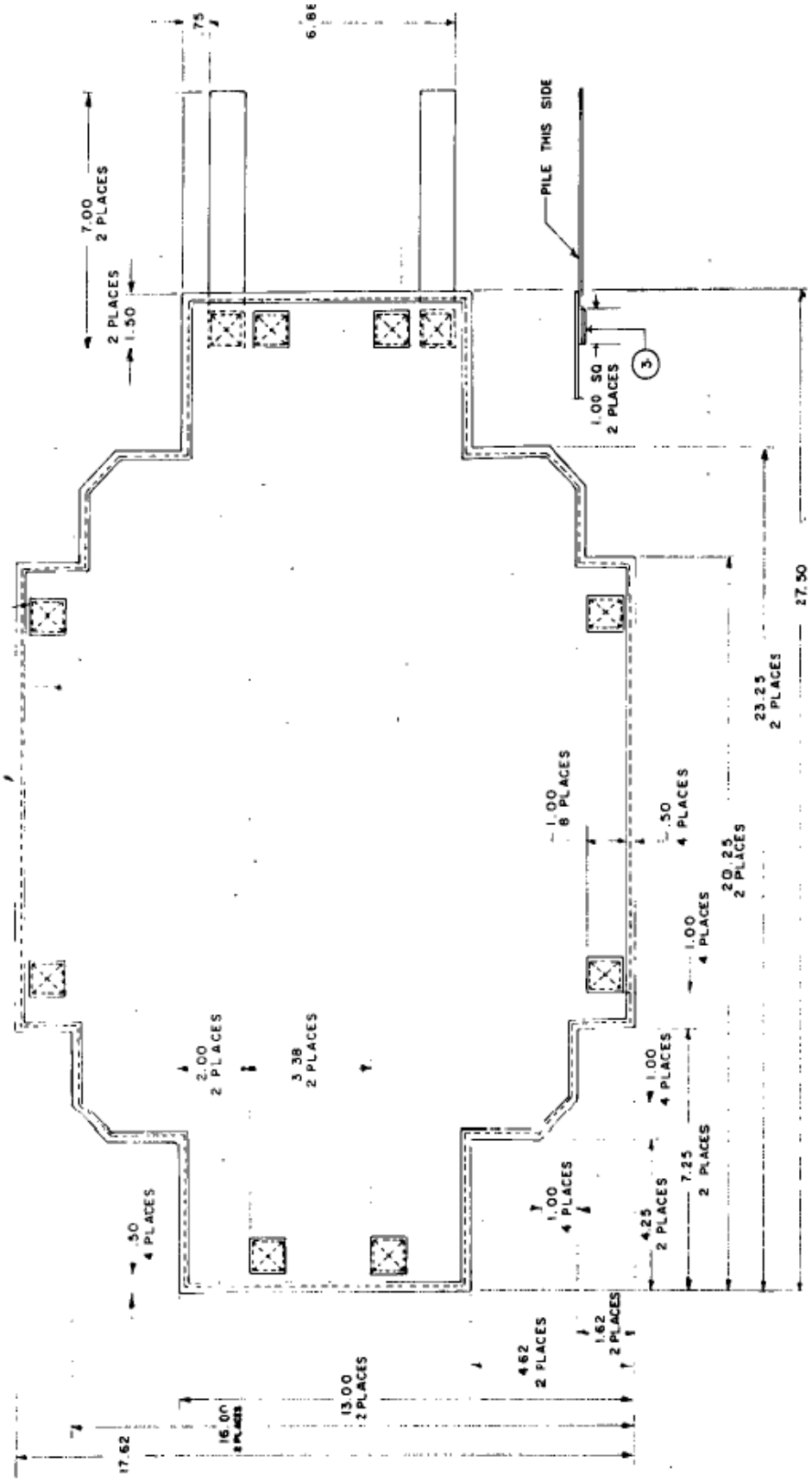
FIND NO.	QTY.	DESCRIPTION AND SPECIFICATION
1	1	PHOSPHOR BRONZE DOWEL, .06±.02 DIA
2	1	BRASS CLAMP WASHER
3	1	PHOSPHOR BRONZE SPRING, 6 TURNS NO.26 GAGE (B & S WIRE) WITH .50 FREE LENGTH AND .20 DIA FLATTENED COIL AT EACH END
4	1	BRASS CLAMP PLATE .62±.02 DIA
5	2	BRASS SCREW, WOOD RD HD, NO.2 BY .50
6	1	BRASS CLAMP NUT .50 DIA(KNURL) TAPPED 6-40 UNF-1B
7	1	BRASS CLAMP SCREW 6-40 UNF-1A

NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. PLACE DOWEL DIAGONALLY OPPOSITE FOUR CORNERS.
3. UNLESS OTHERWISE SPECIFIED TOLERANCES SHALL BE ±.03.

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FIGURE 3. Screw clamp details and arrangement showing dowel location.

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FIND NO	QTY	DESCRIPTION AND SPECIFICATION
1	AR	COTTON DUCK
2	AR	TAPE, BINDING
5	AR	PILE TAPE

NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. TOLERANCES SHALL BE ±.12.

FIGURE 4. Waterproof cover.

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